

DBS

SECTION 1: DDL (DATA DEFINITION LANGUAGE)

DDL COMMANDS:

1. CREATE
2. ALTER
3. DROP
4. TRUNCATE
5. RENAME

QUESTION 1 - CREATE TABLE

```
mysql> CREATE DATABASE bank_db;
Query OK, 1 row affected (0.01 sec)

mysql> USE bank_db;
Database changed
mysql> CREATE TABLE bank_transactions (
  ->     txn_id INT PRIMARY KEY,
  ->     customer_name VARCHAR(50),
  ->     branch_name VARCHAR(50),
  ->     transaction_type VARCHAR(20),
  ->     amount DECIMAL(10,2),
  ->     transaction_date DATE
  -> );
Query OK, 0 rows affected (0.07 sec)

mysql> SHOW TABLES;
+-----+
| Tables_in_bank_db |
+-----+
| bank_transactions |
+-----+
1 row in set (0.01 sec)

mysql> DESCRIBE bank_transactions;
+-----+-----+-----+-----+-----+-----+
| Field          | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| txn_id         | int           | NO   | PRI | NULL    |       |
| customer_name  | varchar(50)   | YES  |     | NULL    |       |
| branch_name    | varchar(50)   | YES  |     | NULL    |       |
| transaction_type | varchar(20)   | YES  |     | NULL    |       |
| amount         | decimal(10,2) | YES  |     | NULL    |       |
| transaction_date | date          | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.01 sec)
```

QUESTION 2 - ALTER TABLE ADD COLUMN

```
mysql> USE bank_db;
Database changed
mysql> ALTER TABLE bank_transactions
    -> ADD account_no VARCHAR(20);
Query OK, 0 rows affected (0.10 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

```
mysql> DESCRIBE bank_transactions;
```

Field	Type	Null	Key	Default	Extra
txn_id	int	NO	PRI	NULL	
customer_name	varchar(50)	YES		NULL	
branch_name	varchar(50)	YES		NULL	
transaction_type	varchar(20)	YES		NULL	
amount	decimal(10,2)	YES		NULL	
transaction_date	date	YES		NULL	
account_no	varchar(20)	YES		NULL	

7 rows in set (0.03 sec)

QUESTION 3 - ALTER TABLE MODIFY COLUMN

```
mysql> USE bank_db;
Database changed
mysql> ALTER TABLE bank_transactions
    -> MODIFY COLUMN customer_name VARCHAR(100);
Query OK, 0 rows affected (0.14 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

```
mysql> DESCRIBE bank_transactions;
```

Field	Type	Null	Key	Default	Extra
txn_id	int	NO	PRI	NULL	
customer_name	varchar(100)	YES		NULL	
branch_name	varchar(50)	YES		NULL	
transaction_type	varchar(20)	YES		NULL	
amount	decimal(10,2)	YES		NULL	
transaction_date	date	YES		NULL	
account_no	varchar(20)	YES		NULL	

7 rows in set (0.01 sec)

QUESTION 4 - RENAME TABLE

```
mysql> USE bank_db;
Database changed
mysql> RENAME TABLE bank_transactions
    -> TO customer_transactions;
Query OK, 0 rows affected (0.09 sec)

mysql> SHOW TABLES;
+-----+
| Tables_in_bank_db |
+-----+
| customer_transactions |
+-----+
1 row in set (0.04 sec)
```

QUESTION 5 - TRUNCATE TABLE

```
mysql> USE bank_db;
Database changed
mysql> TRUNCATE TABLE bank_transactions;
ERROR 1146 (42S02): Table 'bank_db.bank_transactions' doesn't exist
mysql> SHOW TABLES;
+-----+
| Tables_in_bank_db |
+-----+
| customer_transactions |
+-----+
1 row in set (0.01 sec)

mysql> TRUNCATE TABLE customer_transactions;
Query OK, 0 rows affected (0.11 sec)

mysql> ^C
mysql> SELECT * FROM customer_transactions;
Empty set (0.01 sec)
```

QUESTION 6 - DROP TABLE

```
mysql> DROP TABLE bank_backup;
ERROR 1051 (42S02): Unknown table 'bank_db.bank_backup'
mysql> DROP TABLE bank_backup;
ERROR 1051 (42S02): Unknown table 'bank_db.bank_backup'
mysql> CREATE TABLE bank_backup (
  ->     id INT PRIMARY KEY,
  ->     customer_name VARCHAR(100)
  -> );
Query OK, 0 rows affected (0.07 sec)

mysql> SHOW TABLES;
+-----+
| Tables_in_bank_db |
+-----+
| bank_backup       |
| customer_transactions |
+-----+
2 rows in set (0.00 sec)

mysql> DROP TABLE bank_backup;
Query OK, 0 rows affected (0.04 sec)

mysql> SHOW TABLES;
+-----+
| Tables_in_bank_db |
+-----+
| customer_transactions |
+-----+
1 row in set (0.00 sec)

mysql> DROP TABLE IF EXISTS bank_backup;
Query OK, 0 rows affected, 1 warning (0.01 sec)
```

SECTION 2: DML (DATA MANIPULATION LANGUAGE)

DML COMMANDS:

1. INSERT
2. UPDATE
3. DELETE
4. SELECT

STEP 1: INSERT SAMPLE DATA

```
INSERT INTO bank_transactions VALUES
(101, 'Ravi', 'Hyderabad', 'Deposit', 5000, '2024-01-05'),
(102, 'Sita', 'Hyderabad', 'Withdrawal', 2000, '2024-01-06'),
(103, 'Kiran', 'Vijayawada', 'Deposit', 12000, '2024-01-08'),
(104, 'Anil', 'Vizag', 'Deposit', 8000, '2024-01-10'),
(105, 'Priya', 'Hyderabad', 'Withdrawal', 3500, '2024-01-11'),
(106, 'Ramesh', 'Vizag', 'Deposit', 15000, '2024-01-12'),
(107, 'Keerthi', 'Vijayawada', 'Withdrawal', 1000, '2024-01-13'),
(108, 'Rahul', 'Hyderabad', 'Deposit', 9000, '2024-01-14'),
(109, 'Sneha', 'Vizag', 'Withdrawal', 4000, '2024-01-15'),
(110, 'Madhu', 'Vijayawada', 'Deposit', 11000, '2024-01-16');
```

QUESTION 7 - INSERT

```
mysql> USE bank_db;
Database changed
mysql> SHOW TABLES;
+-----+
| Tables_in_bank_db |
+-----+
| customer_transactions |
+-----+
1 row in set (0.00 sec)

mysql> DESC customer_transactions;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| txn_id | int | NO | PRI | NULL | |
| customer_name | varchar(100) | YES | | NULL | |
| branch_name | varchar(50) | YES | | NULL | |
| transaction_type | varchar(20) | YES | | NULL | |
| amount | decimal(10,2) | YES | | NULL | |
| transaction_date | date | YES | | NULL | |
| account_no | varchar(20) | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
7 rows in set (0.00 sec)

mysql> INSERT INTO customer_transactions
-> VALUES
-> (111, 'Venu', 'Vizag', 'Deposit', 7000, '2024-01-18', 'ACC111');
Query OK, 1 row affected (0.09 sec)

mysql> SELECT * FROM customer_transactions;
+-----+-----+-----+-----+-----+-----+-----+
| txn_id | customer_name | branch_name | transaction_type | amount | transaction_date | account_no |
+-----+-----+-----+-----+-----+-----+-----+
| 105 | Priya | Hyderabad | Withdrawal | 5000.00 | 2024-01-11 | ACC105 |
| 111 | Venu | Vizag | Deposit | 7000.00 | 2024-01-18 | ACC111 |
+-----+-----+-----+-----+-----+-----+-----+
2 rows in set (0.01 sec)
```

QUESTION 8 - UPDATE

```
mysql> USE bank_db;
Database changed
mysql> SELECT * FROM customer_transactions;
+-----+-----+-----+-----+-----+-----+-----+
| txn_id | customer_name | branch_name | transaction_type | amount | transaction_date | account_no |
+-----+-----+-----+-----+-----+-----+-----+
| 105 | Priya | Hyderabad | Withdrawal | 5000.00 | 2024-01-11 | ACC105 |
| 111 | Venu | Vizag | Deposit | 7000.00 | 2024-01-18 | ACC111 |
+-----+-----+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)

mysql> UPDATE customer_transactions
-> SET amount = 5000
-> WHERE txn_id = 105;
Query OK, 0 rows affected (0.01 sec)
Rows matched: 1 Changed: 0 Warnings: 0

mysql> SELECT * FROM customer_transactions;
+-----+-----+-----+-----+-----+-----+-----+
| txn_id | customer_name | branch_name | transaction_type | amount | transaction_date | account_no |
+-----+-----+-----+-----+-----+-----+-----+
| 105 | Priya | Hyderabad | Withdrawal | 5000.00 | 2024-01-11 | ACC105 |
| 111 | Venu | Vizag | Deposit | 7000.00 | 2024-01-18 | ACC111 |
+-----+-----+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)
```

QUESTION 9 - DELETE

```
mysql> USE bank_db;
Database changed
mysql> SELECT * FROM customer_transactions;
+-----+-----+-----+-----+-----+-----+-----+
| txn_id | customer_name | branch_name | transaction_type | amount | transaction_date | account_no |
+-----+-----+-----+-----+-----+-----+-----+
| 105 | Priya | Hyderabad | Withdrawal | 5000.00 | 2024-01-11 | ACC105 |
| 111 | Venu | Vizag | Deposit | 7000.00 | 2024-01-18 | ACC111 |
+-----+-----+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)

mysql> DELETE FROM customer_transactions
-> WHERE txn_id = 111;
Query OK, 1 row affected (0.02 sec)

mysql> SELECT * FROM customer_transactions;
+-----+-----+-----+-----+-----+-----+-----+
| txn_id | customer_name | branch_name | transaction_type | amount | transaction_date | account_no |
+-----+-----+-----+-----+-----+-----+-----+
| 105 | Priya | Hyderabad | Withdrawal | 5000.00 | 2024-01-11 | ACC105 |
+-----+-----+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

QUESTION 10 - SELECT

```
mysql> USE bank_db;
Database changed
mysql> SELECT * FROM customer_transactions;
+-----+-----+-----+-----+-----+-----+-----+
| txn_id | customer_name | branch_name | transaction_type | amount | transaction_date | account_no |
+-----+-----+-----+-----+-----+-----+-----+
| 105 | Priya | Hyderabad | Withdrawal | 5000.00 | 2024-01-11 | ACC105 |
+-----+-----+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql> SELECT * FROM customer_transactions;
+-----+-----+-----+-----+-----+-----+-----+
| txn_id | customer_name | branch_name | transaction_type | amount | transaction_date | account_no |
+-----+-----+-----+-----+-----+-----+-----+
| 105 | Priya | Hyderabad | Withdrawal | 5000.00 | 2024-01-11 | ACC105 |
+-----+-----+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

QUESTION 11 - SELECT WITH WHERE

```
mysql> USE bank_db;
Database changed
mysql> SELECT *
    -> FROM customer_transactions
    -> WHERE transaction_type = 'Deposit';
Empty set (0.02 sec)
```

QUESTION 12 - SELECT WITH ORDER BY

```
mysql> USE bank_db;
Database changed
mysql> SELECT *
    -> FROM customer_transactions
    -> ORDER BY amount DESC;
+-----+-----+-----+-----+-----+-----+-----+
| txn_id | customer_name | branch_name | transaction_type | amount | transaction_date | account_no |
+-----+-----+-----+-----+-----+-----+-----+
| 105 | Priya | Hyderabad | Withdrawal | 5000.00 | 2024-01-11 | ACC105 |
+-----+-----+-----+-----+-----+-----+-----+
1 row in set (0.01 sec)
```

SECTION 3: DCL (DATA CONTROL LANGUAGE)

DCL COMMANDS:

1. GRANT
2. REVOKE

QUESTION 13 - GRANT SELECT

```
mysql> USE bank_db;
Database changed
mysql> CREATE USER 'Auditor1'@'localhost' IDENTIFIED BY 'password';
Query OK, 0 rows affected (0.17 sec)

mysql> GRANT SELECT
      -> ON customer_transactions
      -> TO 'Auditor1'@'localhost';
Query OK, 0 rows affected (0.01 sec)

mysql> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.02 sec)
```

QUESTION 14 - GRANT ALL PRIVILEGES

```
mysql> USE bank_db;
Database changed
mysql> GRANT ALL
      -> ON customer_transactions
      -> TO 'BranchManager'@'localhost';
Query OK, 0 rows affected (0.07 sec)

mysql> SHOW GRANTS FOR 'BranchManager'@'localhost';
+-----+
| Grants for BranchManager@localhost |
+-----+
| GRANT USAGE ON *.* TO `BranchManager`@`localhost` |
| GRANT ALL PRIVILEGES ON `bank_db`.`customer_transactions` TO `BranchManager`@`localhost` |
+-----+
2 rows in set (0.01 sec)
```

QUESTION 15 - REVOKE SELECT

```
mysql> USE bank_db;
Database changed
mysql> REVOKE SELECT
      -> ON customer_transactions
      -> FROM 'Auditor1'@'localhost';
Query OK, 0 rows affected (0.02 sec)
```


QUESTION 16 - REVOKE ALL

```
mysql> USE bank_db;
Database changed
mysql> REVOKE ALL
    -> ON customer_transactions
    -> FROM 'BranchManager'@'localhost';
Query OK, 0 rows affected (0.01 sec)
```

SECTION 4: TCL (TRANSACTION CONTROL LANGUAGE)

TCL COMMANDS:

1. COMMIT
2. SAVEPOINT
3. ROLLBACK

QUESTION 17 - COMMIT

```
mysql> USE bank_db;
Database changed
mysql> SELECT * FROM customer_transactions;
+-----+-----+-----+-----+-----+-----+-----+
| txn_id | customer_name | branch_name | transaction_type | amount | transaction_date | account_no |
+-----+-----+-----+-----+-----+-----+-----+
| 105 | Priya | Hyderabad | Withdrawal | 5000.00 | 2024-01-11 | ACC105 |
+-----+-----+-----+-----+-----+-----+-----+
1 row in set (0.02 sec)

mysql> UPDATE customer_transactions
    -> SET amount = 6000
    -> WHERE txn_id = 105;
Query OK, 1 row affected (0.02 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)

mysql> SELECT * FROM customer_transactions;
+-----+-----+-----+-----+-----+-----+-----+
| txn_id | customer_name | branch_name | transaction_type | amount | transaction_date | account_no |
+-----+-----+-----+-----+-----+-----+-----+
| 105 | Priya | Hyderabad | Withdrawal | 6000.00 | 2024-01-11 | ACC105 |
+-----+-----+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

QUESTION 18 - SAVEPOINT

```
mysql> USE bank_db;
Database changed
mysql> SAVEPOINT Before_Update;
Query OK, 0 rows affected (0.00 sec)
```

QUESTION 19 - ROLLBACK TO SAVEPOINT

```
mysql> SAVEPOINT Before_Update;
Query OK, 0 rows affected (0.00 sec)

mysql> UPDATE customer_transactions
    -> SET amount = 99999
    -> WHERE txn_id = 105;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> SELECT * FROM customer_transactions;
+-----+-----+-----+-----+-----+-----+-----+
| txn_id | customer_name | branch_name | transaction_type | amount | transaction_date | account_no |
+-----+-----+-----+-----+-----+-----+-----+
| 105 | Priya | Hyderabad | Withdrawal | 99999.00 | 2024-01-11 | ACC105 |
+-----+-----+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql> ROLLBACK TO Before_Update;
Query OK, 0 rows affected (0.01 sec)

mysql> SELECT * FROM customer_transactions;
+-----+-----+-----+-----+-----+-----+-----+
| txn_id | customer_name | branch_name | transaction_type | amount | transaction_date | account_no |
+-----+-----+-----+-----+-----+-----+-----+
| 105 | Priya | Hyderabad | Withdrawal | 6000.00 | 2024-01-11 | ACC105 |
+-----+-----+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

QUESTION 20 - FULL TCL EXAMPLE

```
mysql> USE bank_db;
Database changed
mysql> SET autocommit = 0;
Query OK, 0 rows affected (0.00 sec)

mysql> SELECT * FROM customer_transactions;
+-----+-----+-----+-----+-----+-----+-----+
| txn_id | customer_name | branch_name | transaction_type | amount | transaction_date | account_no |
+-----+-----+-----+-----+-----+-----+-----+
| 105 | Priya | Hyderabad | Withdrawal | 6000.00 | 2024-01-11 | ACC105 |
+-----+-----+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql> UPDATE customer_transactions
-> SET amount = 7000
-> WHERE txn_id = 105;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> SAVEPOINT SP1;
Query OK, 0 rows affected (0.00 sec)

mysql> UPDATE customer_transactions
-> SET amount = 9000
-> WHERE txn_id = 105;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> SELECT * FROM customer_transactions;
+-----+-----+-----+-----+-----+-----+-----+
| txn_id | customer_name | branch_name | transaction_type | amount | transaction_date | account_no |
+-----+-----+-----+-----+-----+-----+-----+
| 105 | Priya | Hyderabad | Withdrawal | 9000.00 | 2024-01-11 | ACC105 |
+-----+-----+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql> ROLLBACK TO SP1;
Query OK, 0 rows affected (0.00 sec)

mysql> ROLLBACK TO SP1;
Query OK, 0 rows affected (0.00 sec)

mysql> SELECT * FROM customer_transactions;
+-----+-----+-----+-----+-----+-----+-----+
| txn_id | customer_name | branch_name | transaction_type | amount | transaction_date | account_no |
+-----+-----+-----+-----+-----+-----+-----+
| 105 | Priya | Hyderabad | Withdrawal | 7000.00 | 2024-01-11 | ACC105 |
+-----+-----+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql> COMMIT;
Query OK, 0 rows affected (0.01 sec)

mysql> SELECT * FROM customer_transactions;
+-----+-----+-----+-----+-----+-----+-----+
| txn_id | customer_name | branch_name | transaction_type | amount | transaction_date | account_no |
+-----+-----+-----+-----+-----+-----+-----+
| 105 | Priya | Hyderabad | Withdrawal | 7000.00 | 2024-01-11 | ACC105 |
+-----+-----+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)
```